TEXORA · SESSIONAL NOTES · STUDY SHEET

Critical Analysis in Apparels, Report Submission and Presentation

Closing the sessional as an engineer closes a job: analysing the case data critically with a Pareto of its defects, writing the report so another engineer can audit it, and presenting the findings for questioning.

AM-4202 · ADVANCED APPAREL MANUFACTURING-II SESSIONAL · STUDY 7 · LEVEL 4 TERM 2

1. Aim and place of the study

Aim: to perform a critical analysis of the manufacturing case of Study 6 — separating what the data prove from what they merely suggest — and to submit and present the sessional report in the standard engineering format.

"Critical" has its exact meaning here: the analysis must test the case's own numbers, including the comfortable ones. A report that only celebrates 91.2% efficiency and a passed audit is a summary, not an analysis; the analysis begins by asking why DHU ran above target on a line that was supposedly balanced.

2. The defect data, analysed

The 40 defects classified from the case's inspection records (defects, not defective garments — the Pareto counts occurrences) arrange as follows:

DEFECT	COUNT	SHARE %	CUMULATIVE %
Skipped stitch	12	30.0	30.0
Open seam	9	22.5	52.5
Measurement out of tolerance	7	17.5	70.0
Stain / spot	5	12.5	82.5
Puckering	4	10.0	92.5
Other (all remaining types)	3	7.5	100.0

Total		**40**	**100.0**

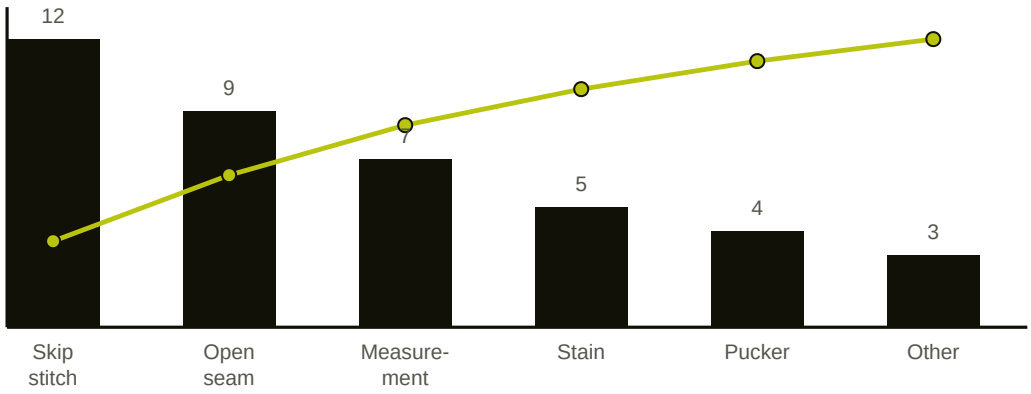


Figure 1 — Pareto chart of the case's defects. Three defect types — skipped stitch, open seam and measurement — carry 70.0% of all defects; any improvement programme that starts elsewhere is misdirected.

3. Reading the analysis critically

- **The vital few.** Skipped stitch and open seam together are 52.5% of defects and both are machine-condition defects: they point at needle state, hook timing and thread control on specific machines (Study 1), not at operator care in general. The corrective action is therefore targeted maintenance plus a needle-change schedule — an action the data can name because the classification was honest.
- **Measurement at 17.5%** on a CAD-graded, CAD-cut order is the finding that should embarrass the plan: it directs attention back to cutting accuracy and to size-wise segregation in sewing, and it is exactly the class of finding a summary-only report would have buried inside a passing audit.
- **What the data cannot say.** The counts carry no time stamps in this record, so the analysis cannot claim whether defects clustered on day one (learning curve) or spread evenly — a limitation to state in the report, with a recommendation that future records log defects by hour and operation. Critical analysis includes auditing one's own evidence.
- **The audit still passed.** Both statements are true at once: the lot met its AQL decision and the process ran dirtier than its target. Holding the two together — shipment decision by sampling, process judgement by DHU and Pareto — is the professional habit this study exists to teach.

4. The report: required structure

SECTION	CONTENT RULE
Title page & contents	Study title, course code, student name/ID, date; contents matching the section numbers actually used.

Aim and theory	The aim in one sentence, then only the theory the work uses — formulas stated before they are applied.
Apparatus / materials	Named machines and materials with the settings that matter (speeds, densities, sizes) — enough for repetition.
Procedure	Numbered, in the order performed; deviations from the planned procedure recorded, not hidden.
Data and calculations	Raw readings first, then the worked calculation with units at every step — the reader must be able to re-derive every result, as in Studies 4–6.
Results and discussion	Results stated plainly, then discussed critically: what they prove, what they suggest, what they cannot show (§3 is the model).
Precautions and conclusion	The precautions actually observed; conclusions numbered and tied to evidence, no new claims appearing for the first time here.

5. The presentation

The presentation defends the report, it does not re-read it. Ten minutes is organised as: the aim and the set-up (2 minutes), the two or three results that matter — here the balance/efficiency pair and the Pareto (5 minutes), the critical findings and recommendations (2 minutes), and one minute of honest limitations. Slides carry the figures (the balancing chart, the Pareto), not paragraphs. Expected viva questions are prepared from the report's own weak points: why efficiency is not 100% on a balanced line, why the audit passed while DHU ran high, what the next experiment would be. A candidate who has read their own report critically has already answered them.

Precautions

- Never adjust data to fit the theory: a discrepancy reported and discussed earns more than a discrepancy concealed, and concealment discovered in viva ends the defence.
- Every number in the presentation must trace to a table in the report; no orphan figures on slides.
- State the teaching-case nature of Study 6 wherever its figures are quoted, so no reader mistakes them for industrial measurements.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. From the Pareto table, what share of defects do the top two types carry, and what kind of cause do they suggest?

2. A fellow student writes: "The lot passed the audit, so quality was satisfactory." Write the two-sentence critical reply.
3. Why must raw readings appear in the report before the calculations?
4. Name two questions you would expect in this study's viva, from the report's own weak points.

Answer key

1. 12 + 9 = 21 of 40 = 52.5%. Both are stitch-formation defects, suggesting machine condition (needle, hook timing, thread control) at specific workstations rather than general workmanship — so maintenance and needle discipline are the corrective actions.
2. "The audit decision concerns this lot against its AQL and was correctly a pass; the process judgement is separate — DHU 6.67 against a target of 5 and a defect Pareto led by machine-condition faults show a process running worse than its standard. Passing a lot is not the same as running a good process."
3. So that every result can be re-derived and audited: calculations on hidden data cannot be checked, and an error in a reading — discovered later — can then be traced to exactly the results it affects.
4. Examples: why the balanced line's efficiency is 91.2% and not 100%; how the lot could pass the AQL audit while DHU exceeded target; what additional data (defects by hour/operation) the next run should record. Any two questions aimed at genuine weak points earn credit.
